

A Function without name is called Anonymous | Lambda Function

```
In [1]: def square(n):  
        return n*n  
  
square(5)  
# i dont want to call square() multiple times
```

Out[1]: 25

```
In [2]: # Lambda  
f = lambda a : a*a  
result = f(5)  
result
```

Out[2]: 25

```
In [4]: f1 = lambda a,b : a-b  
f2 = lambda a,b : a+b  
  
res1 = f1(10,5)  
res2 = f2(2,3)  
  
print(res1)  
print(res2)
```

5

5

```
In [6]: import keyword  
keyword.kwlist
```

```
Out[6]: ['False',
        'None',
        'True',
        'and',
        'as',
        'assert',
        'async',
        'await',
        'break',
        'class',
        'continue',
        'def',
        'del',
        'elif',
        'else',
        'except',
        'finally',
        'for',
        'from',
        'global',
        'if',
        'import',
        'in',
        'is',
        'lambda',
        'nonlocal',
        'not',
        'or',
        'pass',
        'raise',
        'return',
        'try',
        'while',
        'with',
        'yield']
```

1.filter

2.map

3.reduce

```
In [13]: # i want to find the list of even numbers
def is_even(n):
    return n%2 == 0

nums = [2,5,6,7,4,10,25,9]
evens = list(filter(is_even, nums))

print(evens)
```

[2, 6, 4, 10]

```
In [14]: ## i want to find the list of even numbers
```

```
In [15]: def is_odd(n):
          return n%2 != 0

          nums = [2,5,6,7,4,10,25,9]
          odd = list(filter(is_odd, nums))

          print(odd)
```

[5, 7, 25, 9]

```
In [26]: nums = [2,5,6,7,4,10,25,9]
          odd = list(filter(lambda n : n%2 !=0, nums))
          print(odd)
```

[5, 7, 25, 9]

```
In [28]: nums = [2,5,6,7,4,10,25,9]

          evens = list(filter(lambda n : n%2==0, nums))
          odd = list(filter(lambda n: n%2!=0, nums))

          print(evens)
          print(odd)
```

[2, 6, 4, 10]

[5, 7, 25, 9]

- What ever even number I have from the assigned list - I want to double the even number i.e 2 become 4 || 4 become 6 || 6 become 8 - that we will do using map function - this largely we are using in google map reduce programme - we can build using user define & lambda

```
In [29]: # Lets write above function using help of lambda & lambda helps to reduce the li
          nums = [2,5,6,7,4,10,25,9]

          evens = list(filter(lambda n : n%2 ==0, nums))
          odd = list(filter(lambda n : n%2 !=0, nums))

          print(evens)
          print(odd)
```

[2, 6, 4, 10]

[5, 7, 25, 9]

```
In [31]: nums = [2,5,6,7,4,10,25,9]

          evens = list(filter(is_even, nums))
          double = list(map(lambda n : n*2, evens))

          print(evens)
          print(double)
```

[2, 6, 4, 10]

[4, 12, 8, 20]

```
In [32]: nums = [2,5,6,7,4,10,25,9]

          evens = list(filter(is_even, nums))
          double = list(map(lambda n : n*2, evens))
          double_ = list(map(lambda n : n+2, evens))
          double_1 = list(map(lambda n : n-2, evens))

          print(evens)
          print(double)
```

```
print(double_)
print(double_1)
```

```
[2, 6, 4, 10]
[4, 12, 8, 20]
[4, 8, 6, 12]
[0, 4, 2, 8]
```

```
In [34]: nums = [2,5,6,7,4,10,25,9]

evens = list(filter(is_even, nums))

double = list(map(lambda n : n*2, evens))
double_ = list(map(lambda n : n-2, evens))

print(double)
print(double_)
```

```
[4, 12, 8, 20]
[0, 4, 2, 8]
```

- i want to perform reduce now - i want reduce all the values - reduce you can add only 2 values - [2,5,6,7,4,10,25,9] if you sum everything then you will get 56

```
In [45]: from functools import reduce

def add_all(a,b):
    return a+b

nums = [2,5,6,7,4,10,25,9]

evens = list(filter(is_even, nums))
double = list(map(lambda n : n*2, evens))

sums = reduce(add_all, double)
sums
print(double)
print(sums)
```

```
[4, 12, 8, 20]
44
```

```
In [46]: from functools import reduce

nums = [2,5,6,7,4,10,25,9]

evens = list(filter(is_even, nums))
double = list(map(lambda n : n*2, evens))
sums = (reduce(lambda a,b : a + b, double))

print(evens)
print(double)
print(sums)
```

```
[2, 6, 4, 10]
[4, 12, 8, 20]
44
```

```
In [ ]:
```